ALEX CHEN

MACHINE LEARNING & AI RESEARCHER

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EDUCATION

University of California, Berkeley — Bachelor of Science in Electrical Engineering & Computer Science
GPA: 3.92 / 4.0 | Expected Graduation: May 2026
Relevant Coursework: Deep Neural Networks, Natural Language Processing, Computer Vision, Algorithms, Linear Algebra, Probability Theory

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash
Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, OpenCV, NumPy, Pandas, FastAPI
DevOps & MLOps: Docker, Git, Linux, Weights & Biases, MLflow, ONNX, CUDA

EXPERIENCE

Berkeley Artificial Intelligence Research (BAIR) — *Undergraduate Researcher* (Sep 2024 - Present)

- Conduct research on parameter-efficient fine-tuning (LoRA) and vision transformer attention mechanisms under Prof. Darrell.
- Accelerated multi-GPU training throughput by 28% using PyTorch FSDP and mixed-precision fp16 kernels.
- Authored comprehensive ablation benchmarks comparing ViT backbones on downstream classification tasks.

KEY PROJECTS

DiffusionLens: Latent Diffusion Pipeline [Python, PyTorch, Hugging Face, Transformers]

- Built an end-to-end conditional image generation pipeline using PyTorch with custom classifier-free guidance.
- Implemented flash-attention memory optimizations allowing batch size 8 on single 24GB GPU, earning 1.2k+ GitHub stars.

Autonomous Perception Benchmark [Python, OpenCV, Scikit-Learn, Docker]

- Engineered multi-camera object detection and tracking pipeline achieving 42 FPS on edge hardware packaged in Docker.

AWARDS & CERTIFICATIONS

- Winner, CalHacks 2025 AI Track | Dean's Honors List (All semesters)